A GATED-HULL CHARACTERIZATION OF FINITE TWO-DIMENSIONAL PARTIAL CUBES

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ABSTRACT. Chepoi, Knauer, and Philibert proved that, in a finite partial cube of VC-dimension at most two, the gated hull of every isometric cycle is either a disk or a full subdivision of a complete graph. They conjectured the converse. We prove their conjecture by a finite, purely theoretical argument.

The proof is organized around isometric expansions. We first establish structural reductions and an interval-forcing description of gated hulls, together with projection, minimal-fibre, paired-chain, and trace-retraction criteria. The main step is an expansion theorem: if the overlap of an isometric expansion has VC-dimension at least two, then the expanded partial cube contains an isometric cycle whose gated hull has VC-dimension at least three; no connectedness assumption on the overlap is needed.

To prove the expansion theorem, assume conversely that every crossing cycle has a two-dimensional gated hull. The rank-two structure theorem of Chepoi–Knauer–Philibert resolves each such hull into disk cells or full subdivisions. Their traces determine pairwise disjoint coordinate blocks, and the overlap admits an isometric state embedding into a Cartesian product of complete graphs. A mixed cycle in this resolved Hamming graph lifts to an isometric cycle whose gated hull shatters a triple. If no mixed cycle exists, two old coordinates controlled by distinct factors cannot be shattered: the four corresponding convex factor quadrants would themselves form a mixed cycle. This contradiction proves the expansion theorem and, through the hyperplane characterization of VC-dimension, the desired gated-hull characterization.

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1. Introduction

1.1. **Background and significance.** Partial cubes are graphs that admit isometric embeddings into hypercubes. They arose from the graph-addressing problem of Graham and Pollak [9] and from Djoković's metric characterization of hypercube subgraphs [8]. They now form a basic class in metric graph theory, connecting graph distances with set systems, median geometry, interconnection networks, chemical graph theory, media theory, and oriented-matroid geometry; see the survey [2].

Under an irredundant hypercube embedding, the coordinates are precisely the Djoković–Winkler Θ-classes. Hence the vertex set of a partial cube is a binary set family, and its Vapnik–Chervonenkis dimension is intrinsic. This interpretation links partial cubes with bounded-VC concept classes and sample compression. In particular, tope graphs of oriented matroids, affine oriented matroids, complexes of oriented matroids (COMs), and ample classes are partial cubes [3, 11]; completion and compression results exploit this connection [6, 7].

Convex and gated hulls of isometric cycles are central in several structural theories of partial cubes, including netlike, Peano, hypercellular, and COM tope graphs [12, 4, 11]. Convex cycles also underlie zone graphs and Euler-type formulae [10], while even the optimization problem of finding small convex generating sets is nontrivial [1].

Chepoi, Knauer, and Philibert [5] developed a detailed theory of partial cubes of VC-dimension at most two. Such graphs are obtained from gated cycles and gated full subdivisions of complete graphs by suitable amalgamations, and they admit ample completions without increasing the VC-dimension. Their cycle-hull theorem gives the following local property: the gated hull of every isometric cycle is a disk, that is, a region graph of a pseudoline arrangement, or a full subdivision of a complete graph. They asked whether the converse is true.

Conjecture 1.1 (Chepoi–Knauer–Philibert; Conjecture 46)**.** *If the gated hull of every isometric cycle of a finite partial cube is a disk or a full subdivision of a complete graph, then the partial cube has VC-dimension at most two.*

Theorem 1.2 (Main theorem)**.** *Conjecture 1.1 is true. Equivalently, a finite partial cube has VC-dimension at most two if and only if the gated hull of every isometric cycle is a disk or a full subdivision of a complete graph.*

The statement is a local-to-global theorem. A partial cube has VC-dimension at least three exactly when three Θ-classes are shattered, equivalently when Q_3 occurs as a partial-cube minor. The main theorem says that this global obstruction is always visible inside the gated hull of a single isometric cycle. Thus the rank-two geometry of pseudoline arrangements and full subdivisions is not merely necessary but also sufficient.

1.2. **Related work.** The modern theory of partial cubes uses gated amalgams, Cartesian products, hyperplanes, zone graphs, and partial-cube minors. These methods occur in the study of cellular, netlike, Peano, and hypercellular graphs and in the graph theory of COMs [12, 3, 4, 11]. The two-dimensional theory in [5] combines these metric tools with affine oriented matroids, pseudoline arrangements, and ample completions. Further completion and sample-compression results appear in [6, 7].

Recent contributions by Xie, Feng, and Xu illustrate continuing work on symmetry, convex cycles, and Θ-class interactions in partial cubes, including transposition-generated Cayley graphs, regular 2-arc-transitive partial cubes, cube polynomials and crossing graphs, and regular partial cubes with prescribed convex-cycle lengths [13, 14, 15, 16]. These results are complementary to the gated-hull characterization proved here.

1.3. **Proof architecture.** The argument has four layers.

(1) We establish structural reductions. High-dimensional isometric antipodal subgraphs, finite COM tope graphs, proper gated separators, and high-dimensional gated Cartesian products cannot occur in a minimal counterexample.
(2) We characterize gated hulls by an interval-forcing operator and derive coordinate, projection, minimal-fibre, paired-chain, and trace retraction criteria.
(3) For an isometric expansion $H = G^e$ with overlap F, assume that no isometric cycle has a gated hull of VC-dimension at least three. Every crossing cycle hull is then a two-dimensional partial cube. The theorem of Chepoi–Knauer–Philibert resolves it as a disk or a full subdivision. Resolving these rank-two cells produces a transition graph that embeds isometrically into a product of complete graphs.
(4) A mixed cycle in the resolved Hamming graph lifts to an isometric cycle whose gated hull shatters a triple. If there is no mixed cycle, a shattered pair of old coordinates defines two crossing convex factor splits, and those splits themselves force a mixed cycle. Hence every overlap of VC-dimension at least two yields the required cycle. Applying this expansion theorem to a high-dimensional hyperplane proves the main theorem.

Every step of the proof is finite and combinatorial. No enumeration or computer-assisted classification is used in the main theorem.

2. Preliminaries

All graphs are finite, connected, and simple. Distances and intervals are denoted by

$$d_G(u, v), \qquad I_G(u, v) = \{x : d_G(u, x) + d_G(x, v) = d_G(u, v)\}.$$

A vertex set A is *convex* if $I_G(u, v) \subseteq A$ for all $u, v \in A$. It is *gated* if every vertex $x \notin A$ has a unique vertex $g \in A$ such that

$$d_G(x, a) = d_G(x, g) + d_G(g, a) \qquad (a \in A).$$

The vertex g is the gate of x in A. Gated sets are convex, intersections of gated sets are gated, and gatedness is transitive. For $S \subseteq V(G)$, write $\mathrm{gh}_G(S)$ for the smallest gated subgraph containing S.

Let G be a partial cube with irredundant coordinates indexed by a set Ω. For $X \subseteq \Omega$, contraction of all coordinates outside X is denoted by π_X. A coordinate set X is shattered by G precisely when $\pi_X(G) = Q_X$. We use the standard facts that convex subgraphs of partial cubes are restrictions, contractions preserve partial cubes, and contractions send gated subgraphs to gated subgraphs [5].

For vertices x, y we write

$$\Delta(x, y) = \{q \in \Omega : x_q \neq y_q\}.$$

Because the embedding is isometric,

$$d_G(x, y) = |\Delta(x, y)|, \tag{1}$$

and the interval has the coordinate description

$$z \in I_G(x, y) \iff \Delta(x, z) \subseteq \Delta(x, y). \tag{2}$$

Equivalently, after translating x to $\varnothing$ and writing $y = R$, the interval $I_G(x, y)$ is the part of the Boolean interval $[\varnothing, R]$ that belongs to G.

Lemma 2.1 (Geodesic label rule)**.** *A path in a partial cube is geodesic if and only if it uses no Θ-class more than once.*

Proof. If a path repeats a coordinate, its length is strictly larger than the Hamming distance of its endpoints. Conversely, if its edge labels are pairwise distinct, every traversed coordinate changes the endpoint label, so the path length equals the Hamming distance in (1). □

Lemma 2.2 (Label word of an isometric cycle)**.** *Let $C = (v_0, v_1, \ldots, v_{2k-1}, v_0)$ be an isometric cycle; necessarily $k \geq 2$. After choosing an orientation and a starting vertex, its edge-label word is*

$$a_1 a_2 \cdots a_k a_1 a_2 \cdots a_k,$$

where $a_1, \ldots, a_k$ are pairwise distinct. Conversely, for $k \geq 2$, a closed walk with this label word is an isometric $2k$-cycle.

Proof. Both v_0–v_k arcs are geodesics, so the labels on each arc are pairwise distinct and the two arcs use the same set of coordinates. Write the first labels as $a_1, \ldots, a_k$ and the second, read from v_k to v_0, as $b_1, \ldots, b_k$. Let $A_i = \{a_1, \ldots, a_i\}$ and $B_i = \{b_1, \ldots, b_i\}$, and put $L = \{a_1, \ldots, a_k\} = \{b_1, \ldots, b_k\}$. The vertices reached after i edges on the two oppositely oriented half-cycles are antipodal on C; their Hamming distance is

$$|A_i \cap B_i| + |L \setminus (A_i \cup B_i)| = k - |A_i \triangle B_i|.$$

Both arcs between them have length k. Isometry therefore forces this distance to be k, hence $A_i = B_i$ for every i. Comparing consecutive prefixes gives $a_i = b_i$.

Conversely, every cyclic subwalk of length at most k in $a_1 \cdots a_k a_1 \cdots a_k$ has pairwise distinct labels and is geodesic by Theorem 2.1. If two vertices of the closed walk coincided before the final return, one of the two nonempty arcs between them would have length at most k and would be a geodesic of positive length, a contradiction. Thus the walk is a simple $2k$-cycle, and its shorter arc between any two vertices is geodesic. □

Lemma 2.3 (Nonexpansiveness of gate maps)**.** *If M is gated in a graph G and p_M is its gate map, then*

$$d_G(p_M(x), p_M(y)) \leq d_G(x, y) \qquad (x, y \in V(G)).$$

In particular, adjacent vertices have equal or adjacent gates.

Proof. It is enough to consider adjacent x, y. Put $g = p_M(x)$ and $h = p_M(y)$. The two gate identities and the triangle inequality give

$$d(x, g) + d(g, h) = d(x, h) \leq 1 + d(y, h),$$

$$d(y, h) + d(g, h) = d(y, g) \leq 1 + d(x, g).$$

Adding yields $2d(g, h) \leq 2$. The general statement follows along a shortest x–y path. □

Lemma 2.4 (Coordinate preservation by gates)**.** *Let M be gated in a partial cube K, and let p_M be its gate map. If M crosses a coordinate q, then*

$$p_M(x)_q = x_q \qquad (x \in V(K)).$$

Proof. Put $g = p_M(x)$ and suppose $g_q \neq x_q$. Since M crosses q, choose $m \in M$ with $m_q = x_q$. The coordinate q contributes to both $d(x, g)$ and $d(g, m)$ but not to $d(x, m)$. Therefore

$$d(x, g) + d(g, m) \geq d(x, m) + 2,$$

contrary to the gate identity $d(x, m) = d(x, g) + d(g, m)$. □

Lemma 2.5 (Locality of gated hulls)**.** *If H is gated in G and $S \subseteq V(H)$, then*

$$\mathrm{gh}_H(S) = \mathrm{gh}_G(S).$$

Proof. Let $L = \mathrm{gh}_G(S)$. Since H is gated in G and contains S, minimality gives $L \subseteq H$. Thus L is gated in H, so $\mathrm{gh}_H(S) \subseteq L$.

Conversely, $\mathrm{gh}_H(S)$ is gated in H, and gatedness is transitive. Hence $\mathrm{gh}_H(S)$ is gated in G and contains S, so $L \subseteq \mathrm{gh}_H(S)$. □

Lemma 2.6 (VC-dimension of a Cartesian product)**.** *For finite partial cubes A and B,*

$$\mathrm{VCdim}(A\Box B) = \mathrm{VCdim}(A) + \mathrm{VCdim}(B).$$

Proof. The Θ-classes of $A\Box B$ split into the classes inherited from A and those inherited from B. If X_A and X_B are shattered in the factors, then $X_A \cup X_B$ is shattered in the product.

Conversely, if X is shattered in $A\Box B$, projection to each factor shows that the classes of X belonging to that factor are shattered there. Thus $|X| \leq \mathrm{VCdim}(A) + \mathrm{VCdim}(B)$. □

Lemma 2.7. *Every edge of a connected bipartite graph is gated. Products of gated subgraphs are gated in the Cartesian product.*

Proof. For an edge uv and any vertex x, bipartiteness implies that $d(x, u)$ and $d(x, v)$ differ by one. The closer endpoint is the gate of x in uv. The product assertion follows from additivity of distances in Cartesian products. □

A connected partial cube of VC-dimension at most one is a tree; this follows, for example, from the virtual-isometric-tree characterization in [5, Proposition 9]. Therefore every partial cube containing a cycle has VC-dimension at least two.

Definition 2.8. A partial cube H is *gated-minimal non-two-dimensional* if $\mathrm{VCdim}(H) \geq 3$ and every proper gated subgraph of H has VC-dimension at most two.

Every finite non-two-dimensional partial cube contains such a gated subgraph: choose one minimal by inclusion. By Theorem 2.5, to prove Conjecture 1.1 for finite graphs it is enough to show that every gated-minimal non-two-dimensional partial cube is the gated hull of one isometric cycle.

3. Antipodal subgraphs and COMs

A partial cube H is *antipodal* if every vertex x has an antipode $-x$ satisfying $H = \mathrm{conv}_H(x, -x)$. Knauer and Marc proved that every finite antipodal partial cube contains an isometric cycle whose convex hull is the whole graph [11, Lemma 2.15].

Lemma 3.1 (Convex generation in an isometric supergraph)**.** *If H is an isometric subgraph of a graph G and $S \subseteq V(H)$, then*

$$\mathrm{conv}_H(S) \subseteq \mathrm{conv}_G(S).$$

Proof. Construct the convex hull by iterated interval closure. Since H is isometric in G,

$$I_H(u, v) \subseteq I_G(u, v) \qquad (u, v \in V(H)).$$

An induction over the interval-closure iterations gives the result. □

Theorem 3.2 (Isometric antipodal obstruction)**.** *Let G be a finite partial cube. If G contains an isometric antipodal subgraph H with* $\mathrm{VCdim}(H) \geq 3$*, then G contains an isometric cycle C satisfying*

$$H \subseteq \mathrm{conv}_G(C) \subseteq \mathrm{gh}_G(C).$$

In particular, $\mathrm{VCdim}(\mathrm{gh}_G(C)) \geq 3$.

Proof. Choose, by [11, Lemma 2.15], an isometric cycle $C \subseteq H$ with $\mathrm{conv}_H(C) = H$. Since H is isometric in G, the cycle C is also isometric in G. By Theorem 3.1,

$$H = \mathrm{conv}_H(C) \subseteq \mathrm{conv}_G(C).$$

Every gated subgraph is convex, so $\mathrm{conv}_G(C) \subseteq \mathrm{gh}_G(C)$. VC-dimension is monotone under enlarging the vertex family, and therefore $\mathrm{VCdim}(\mathrm{gh}_G(C)) \geq \mathrm{VCdim}(H) \geq 3$. □

Corollary 3.3. *Conjecture 1.1 holds for finite antipodal partial cubes.*

Proof. If an antipodal partial cube G has VC-dimension at least three, apply Theorem 3.2 with $H = G$. □

We next use two results about complexes of oriented matroids. Tope graphs of COMs are partial cubes in which every antipodal subgraph is gated [11]. Moreover, the VC-dimension of a finite COM is attained by the tope graph of one of its faces, and each face tope graph is antipodal [7, Lemma 5].

Theorem 3.4. *Conjecture 1.1 holds for tope graphs of finite complexes of oriented matroids. Consequently, it holds for finite oriented matroids, affine oriented matroids, and ample (lopsided) partial cubes.*

Proof. Let G be the tope graph of a finite COM with $\mathrm{VCdim}(G) \geq 3$. Choose a face whose tope graph H satisfies $\mathrm{VCdim}(H) = \mathrm{VCdim}(G)$. The graph H is antipodal and isometric in G; indeed it is gated. Apply Theorem 3.2. □

Remark 3.5. The earlier antipodal reduction required H to be gated in G in order to conclude $\mathrm{gh}_G(C) = H$. The strengthened statement above does not claim equality; the inclusion $H \subseteq \mathrm{gh}_G(C)$ is sufficient to force VC-dimension at least three.

4. Gated amalgams and separators

We now prove a general VC-dimension formula for an intersecting union of gated subgraphs.

Theorem 4.1 (Gated-union formula)**.** *Let G be a finite partial cube and suppose*

$$G = G_1 \cup G_2,$$

where G_1 and G_2 are gated subgraphs with $G_1 \cap G_2 \neq \varnothing$. Then

$$\mathrm{VCdim}(G) = \max\{\mathrm{VCdim}(G_1), \mathrm{VCdim}(G_2)\}.$$

Proof. The lower bound is immediate. For the reverse inequality, let X be shattered by G. Contract every Θ-class outside X. Then $\pi_X(G) = Q_X$. Contractions preserve gated subgraphs, so

$$A := \pi_X(G_1), \qquad B := \pi_X(G_2)$$

are gated subgraphs of Q_X, they intersect, and $A \cup B = Q_X$.

A gated subgraph of a hypercube is convex, and every convex subgraph of a hypercube is a subcube. Pick $z \in A \cap B$. If both A and B were proper subcubes, each would fix at least one coordinate to its value at z. The antipode $\bar{z}$ of z in Q_X would then belong to neither A nor B, contradicting $A \cup B = Q_X$. Hence one of A, B equals Q_X, so X is shattered by G_1 or by G_2. □

Lemma 4.2 (Attaching components to a gated separator)**.** *Let S be a gated subgraph of a connected graph G. If $\mathcal{C}$ is any collection of connected components of $G - S$, then*

$$K := S \cup \bigcup_{C \in \mathcal{C}} V(C)$$

is gated in G.

Proof. Only vertices in components not selected for K require gates. Let x be such a vertex and let g be its gate in S. For any $y \in K$, every x–y path meets S. Hence

$$d(x, y) = \min_{s \in S}\big(d(x, s) + d(s, y)\big).$$

Since g is the gate of x in S, $d(x, s) = d(x, g) + d(g, s)$ for all $s \in S$. Therefore

$$d(x, y) = d(x, g) + \min_{s \in S}\big(d(g, s) + d(s, y)\big) = d(x, g) + d(g, y),$$

where the last equality follows by taking $s = g$ and using the triangle inequality. Thus g is the gate of x in K. □

Corollary 4.3 (No proper gated separator)**.** *If H is gated-minimal non-two-dimensional, then $H - S$ is connected for every proper gated subgraph $S \subsetneq H$.*

Proof. Suppose $H - S$ has at least two components. Partition its components into two nonempty collections and adjoin S to each collection. By Theorem 4.2, this gives two proper gated subgraphs H_1, H_2 with nonempty intersection and $H = H_1 \cup H_2$. Minimality gives $\mathrm{VCdim}(H_i) \leq 2$, while Theorem 4.1 gives $\mathrm{VCdim}(H) \leq 2$, a contradiction. □

Corollary 4.4. *Every gated-minimal non-two-dimensional partial cube is 2-connected. Moreover, deleting the two endpoints of any edge leaves the graph connected.*

Proof. Vertices and edges are gated in a connected bipartite graph by Theorem 2.7. A cut vertex or an adjacent two-vertex cut would therefore be a proper gated separator, contrary to Theorem 4.3. □

5. Cartesian-product obstructions

The previous product reduction dealt with a Cartesian-decomposable minimal core. The next theorem removes minimality and requires only a high-dimensional gated product subgraph.

Theorem 5.1 (Gated product obstruction)**.** *Let G be a finite partial cube containing a gated subgraph*

$$H = A \square B,$$

where both factors are nontrivial and $\mathrm{VCdim}(H) \geq 3$. *Then G contains an isometric cycle C such that*

$$\mathrm{VCdim}(\mathrm{gh}_G(C)) \geq 3.$$

Proof. By Theorem 2.6, after interchanging the factors if necessary, $\mathrm{VCdim}(B) \geq 2$. Hence B is not a tree and contains an isometric cycle D. Put

$$B_0 := \mathrm{gh}_B(D).$$

The graph B_0 contains a cycle, so $\mathrm{VCdim}(B_0) \geq 2$.

Choose an edge xy of A. By Theorem 2.7,

$$H_0 := xy \square B_0$$

is gated in H, and hence in G. Write

$$D = (v_0, v_1, \ldots, v_{2k-1}, v_0)$$

with v_0, v_k opposite on D. In the two B_0-layers indexed by x, y, define the belt cycle

$$\begin{aligned} C = {} & (v_0, x), (v_1, x), \ldots, (v_k, x), (v_k, y), \\ & (v_{k+1}, y), \ldots, (v_{2k-1}, y), (v_0, y), (v_0, x). \end{aligned}$$

If the first half of D has pairwise distinct Θ-labels $a_1, \ldots, a_k$ and the product edge has label e, then the label word of C is

$$a_1 \cdots a_k e \, a_1 \cdots a_k e.$$

Every subpath of length at most $k+1$ uses each label at most once and is geodesic; hence C is isometric.

Let $M = \mathrm{gh}_{H_0}(C)$. Convexity of M and the two opposite endpoints in each layer imply that M contains both copies of D. Each layer $B_0 \square \{x\}$ and $B_0 \square \{y\}$ is gated in H_0. By Theorem 2.5 and the definition of B_0, the gated hull of the copy of D in either layer is the whole layer. Hence $M = H_0$. Another use of Theorem 2.5 yields $\mathrm{gh}_G(C) = H_0$. Finally,

$$\mathrm{VCdim}(H_0) = 1 + \mathrm{VCdim}(B_0) \geq 3.$$

□

Corollary 5.2. *A finite counterexample to Conjecture 1.1, chosen gated-minimal, must be Cartesian-prime. More generally, a graph satisfying the hypothesis of Conjecture 1.1 contains no gated Cartesian-product subgraph of VC-dimension at least three with both factors nontrivial.*

For completeness we record the sharper factor structure from the earlier minimal-core argument.

Proposition 5.3 (Factor structure of a decomposable minimal core)**.** *Let H be gated-minimal non-two-dimensional and suppose $H = A\Box B$ with both factors nontrivial. After interchanging factors,*

$$A \cong K_2, \qquad \mathrm{VCdim}(B) = 2,$$

and every proper gated subgraph of B has VC-dimension at most one.

Proof. Put $a = \mathrm{VCdim}(A)$ and $b = \mathrm{VCdim}(B)$. The factor layers are proper gated subgraphs, so $a, b \le 2$. If both factors have more than two vertices, choose edges $e_A \subseteq A$ and $e_B \subseteq B$. The proper gated subgraphs $A\Box e_B$ and $e_A\Box B$ have VC-dimensions $a+1$ and $b+1$, forcing $a, b \le 1$, contrary to $a + b = \mathrm{VCdim}(H) \ge 3$. Thus one factor is K_2, say $A = K_2$. Then $a = 1$ and $b = 2$.

If L is a proper gated subgraph of B, then $K_2\Box L$ is a proper gated subgraph of H. Hence $1 + \mathrm{VCdim}(L) \le 2$. □

6. Gated hulls as interval-forcing closures

We now give a characterization that applies to every finite connected graph, not only to partial cubes.

For a nonempty set $S \subseteq V(G)$ and a vertex $x \in V(G)$ define

$$J_S(x) := \bigcap_{s\in S} I_G(x, s)$$

and the forcing operator

$$\Phi_G(S) := S \cup \{x \in V(G) \setminus S : J_S(x) = \{x\}\}.$$

Theorem 6.1 (Forcing characterization of gated sets)**.** *Let G be a finite connected graph and $\varnothing \ne S \subseteq V(G)$. Then*

$$S \text{ is gated} \iff \Phi_G(S) = S.$$

Moreover, if

$$S_0 = S, \qquad S_{i+1} = \Phi_G(S_i),$$

then the stable set of this iteration is exactly $\mathrm{gh}_G(S)$.

Proof. Suppose first that S is gated. For $x \notin S$, let $g \in S$ be its gate. Then $g \in I_G(x, s)$ for every $s \in S$, so $g \in J_S(x)$. Since $g \ne x$, $J_S(x) \ne \{x\}$, and $\Phi_G(S) = S$.

Conversely, assume $\Phi_G(S) = S$ and fix $x_0 \notin S$. Since $J_S(x_0) \ne \{x_0\}$, choose

$$x_1 \in J_S(x_0) \setminus \{x_0\}.$$

If $x_1 \notin S$, continue. Fix $s_0 \in S$. Since $x_{i+1} \in I_G(x_i, s_0)$ and $x_{i+1} \ne x_i$,

$$d(x_{i+1}, s_0) < d(x_i, s_0).$$

The process therefore terminates at some $g \in S$. For every $s \in S$, repeated interval equalities give

$$d(x_0, s) = d(x_0, g) + d(g, s).$$

Thus g is a gate. It is unique: if $g, h \in S$ are both gates, applying the gate equalities with $s = h$ and $s = g$ forces $d(g, h) = 0$.

It remains to identify the stable set. Let K be any gated set containing S. If $J_S(x) = \{x\}$ and $x \notin K$, the gate g of x in K lies in $I_G(x, s)$ for every $s \in S$, hence $g = x$, a contradiction.

Therefore $\Phi_G(S) \subseteq K$. Inductively every iterate S_i is contained in every gated supergraph of S. Since the finite iteration stabilizes at a fixed point, that stable set is gated by the first part and is the smallest gated supergraph of S. □

We now specialize the forcing characterization to the fixed isometric embedding of a partial cube G in Q_Ω.

Proposition 6.2 (Coordinate form of forcing). *For $S \neq \varnothing$ and $x \in V(G)$, set*

$$A_S(x) := \bigcap_{s \in S} \Delta(x, s).$$

Then

$$J_S(x) = \{y \in V(G) : \Delta(x, y) \subseteq A_S(x)\}.$$

Consequently, x is forced by S exactly when the coordinate subcube

$$\{x \triangle B : B \subseteq A_S(x)\}$$

contains no other vertex of G.

Proof. Because the embedding is isometric,

$$y \in I_G(x, s) \iff \Delta(x, y) \subseteq \Delta(x, s).$$

Intersecting over $s \in S$ proves the formula. □

Let $\mathrm{Box}(S)$ be the smallest coordinate subcube of Q_Ω containing S: coordinates constant on S are fixed and all other coordinates are free.

Corollary 6.3 (Subcube filling). *For every nonempty $S \subseteq V(G)$,*

$$V(G) \cap \mathrm{Box}(S) \subseteq \Phi_G(S).$$

In particular, if S crosses every used coordinate of G, then

$$\Phi_G(S) = V(G)$$

and $\mathrm{gh}_G(S) = G$.

Proof. Let $x \in V(G) \cap \mathrm{Box}(S)$. On every coordinate, some vertex of S agrees with x: this is automatic on coordinates fixed by S, and follows from the occurrence of both signs on free coordinates. Hence $A_S(x) = \varnothing$. By Theorem 6.2, $J_S(x) = \{x\}$. □

7. Cycle projection and minimal fibers

Let C be an isometric cycle in a partial cube $G \subseteq Q_\Omega$, and let

$$X := \Theta(C)$$

be the set of coordinates used by C. Every coordinate outside X is constant on C.

Theorem 7.1 (Cycle-projection theorem). *Let $M = \mathrm{gh}_G(C)$. Then*

$$\pi_X(M) = \pi_X(G).$$

Proof. Contractions preserve gated subgraphs, so $\pi_X(M)$ is gated, hence convex, in $\pi_X(G)$. The isometric cycle C has two opposite vertices u, v that differ in every coordinate of X. For every $z \in \pi_X(G)$, isometricity in Q_X gives

$$d(u, z) + d(z, v) = |X| = d(u, v).$$

Thus $z \in I_{\pi_X(G)}(u, v)$. Since $\pi_X(M)$ is convex and contains u, v, it contains all of $\pi_X(G)$. □

Corollary 7.2 (Direct projection certificate). *If the coordinate set $X = \Theta(C)$ contains a triple shattered by G, then*

$$\mathrm{VCdim}(\mathrm{gh}_G(C)) \geq 3.$$

More generally,

$$\mathrm{VCdim}(\mathrm{gh}_G(C)) \geq \mathrm{VCdim}(\pi_X(G)).$$

Proof. By Theorem 7.1, $\pi_X(G)$ is a contraction of $\mathrm{gh}_G(C)$. Any cube minor of the contraction is therefore a partial-cube minor of $\mathrm{gh}_G(C)$. □

When direct projection does not already expose a shattered triple, the following fiber description can still force vertices into the hull. Let b_r be the constant value of C on each $r \in \Omega \setminus X$, and define the external support

$$A_C(x) := \{r \in \Omega \setminus X : x_r \neq b_r\}.$$

Vertices having the same X-coordinates form an X-fiber.

Theorem 7.3 (Minimal-fiber theorem)**.** *For every $x \in V(G)$,*

$$\bigcap_{c \in V(C)} I_G(x, c) = \{y \in V(G) : \pi_X(y) = \pi_X(x),\ A_C(y) \subseteq A_C(x)\}.$$

Consequently, the first forcing set $\Phi_G(V(C))$ consists exactly of the vertices whose external supports are inclusion-minimal within their X-fibers.

Proof. Apply Theorem 6.2. For $q \in X$, the cycle contains a vertex whose q-coordinate agrees with x_q, so any vertex in the intersection of all intervals must agree with x on q. Thus its X-projection equals that of x.

For $r \notin X$, all cycle vertices have coordinate b_r. If $x_r = b_r$, the coordinate cannot change inside the interval intersection. If $x_r \neq b_r$, it may either remain x_r or change to b_r. This is precisely the condition $A_C(y) \subseteq A_C(x)$. □

Corollary 7.4 (Minimal-fiber whole-hull certificate)**.** *Let F_C be the set of fiberwise inclusion-minimal vertices from Theorem 7.3. If F_C crosses every used coordinate of G, then*

$$\mathrm{gh}_G(C) = G.$$

Proof. By Theorem 7.3, $F_C \subseteq \Phi_G(V(C)) \subseteq \mathrm{gh}_G(C)$. Since F_C crosses every coordinate, Theorem 6.3 shows that the next forcing iteration is all of $V(G)$. □

8. Connected high-dimensional expansion overlaps

We now isolate a large class of isometric expansions that cannot occur in a counterexample. Recall that an *isometric cover* of a connected graph G is a triple (G_1, G_0, G_2) such that G_1, G_2 are isometric subgraphs of G, their vertices and edges cover G, and G_0 is the subgraph induced by $V(G_1) \cap V(G_2)$. The associated isometric expansion $\widetilde{G}$ duplicates each vertex of G_0 and joins its two copies by an edge in a new Θ-class E_e; see [5]. Contracting E_e recovers G.

The overlap G_0 need not be an isometric subgraph of G. Its VC-dimension is nevertheless defined through its vertex set in the ambient hypercube. This distinction is essential in the proof below.

Theorem 8.1 (Connected-overlap obstruction)**.** *Let $\widetilde{G}$ be an isometric expansion of a partial cube G with respect to an isometric cover (G_1, G_0, G_2), and let e be the new coordinate. If G_0 is connected and*

$$\mathrm{VCdim}(G_0) \geq 2,$$

then $\widetilde{G}$ contains an isometric cycle C such that

$$\mathrm{VCdim}(\mathrm{gh}_{\widetilde{G}}(C)) \geq 3.$$

Proof. We distinguish according to whether G_0 is isometric in G.

Case 1: G_0 is isometric in G. Then G_0 is a connected partial cube of VC-dimension at least two, hence it contains an isometric cycle

$$D = (v_0, v_1, \ldots, v_{2k-1}, v_0).$$

Let v_0, v_k be opposite on D. In the two expansion layers define the belt cycle

$$C = v_0^-, v_1^-, \dots, v_k^-, v_k^+,$$
$$v_{k+1}^+, \dots, v_{2k-1}^+, v_0^+, v_0^-.$$

If one half of D has labels $q_1, \dots, q_k$, then the label word of C is

$$q_1 \cdots q_k e\, q_1 \cdots q_k e.$$

Thus every subpath of length at most $k+1$ uses each Θ-class at most once, and C is isometric.

Let $M = \mathrm{gh}_{\widetilde{G}}(C)$. Since M is convex and contains opposite vertices of each copy of D, it contains both copies D^- and D^+. Hence M contains the vertex set $V(D) \times \{0,1\}$. The cycle D has VC-dimension two, so some pair of its coordinates is shattered. Together with e, the corresponding triple is shattered by M. Therefore $\mathrm{VCdim}(M) \geq 3$.

Case 2: G_0 is not isometric in G. Write d_0 for the internal graph distance in G_0. Choose $u, v \in V(G_0)$ with

$$d_0(u, v) > d_G(u, v)$$

so that $d := d_G(u, v)$ is minimum. Because G_1 and G_2 are isometric in G, choose u–v geodesics

$$P_1 \subseteq G_1, \qquad P_2 \subseteq G_2$$

of length d.

No internal vertex of either P_i lies in G_0. Indeed, if an internal vertex $z \in G_0$ occurred, then both $d_G(u, z)$ and $d_G(z, v)$ would be smaller than d. By minimality of d, these two shorter pairs have the same distances in G_0 and in G, whence

$$d_0(u, v) \leq d_0(u, z) + d_0(z, v) = d,$$

a contradiction. The two geodesics also have no common internal vertex, since any such vertex would belong to $G_1 \cap G_2 = G_0$. Hence $C_0 = P_1 \cup P_2$ is a simple cycle.

We claim that C_0 is isometric. Translate the hypercube coordinates so that $u = \varnothing$ and $v = R$ with $|R| = d$. Every vertex of $P_1 \cup P_2$ is then a subset of R. Take $x \in P_1$ and $y \in P_2$, represented by $A, B \subseteq R$. The two x–y arcs of C_0 have lengths

$$|A| + |B| \qquad \text{and} \qquad 2d - |A| - |B|,$$

while $d_G(x, y) = |A \triangle B|$. If $A \cap B = \varnothing$, the arc through u is geodesic; if $A \cup B = R$, the arc through v is geodesic.

Suppose instead that $A \cap B \neq \varnothing$ and $A \cup B \neq R$. Any x–y path must meet G_0, because there is no edge joining $G_1 \setminus G_0$ to $G_2 \setminus G_0$. Choose an x–y geodesic and a vertex $z \in G_0$ on it. If $Z \subseteq R$ is the coordinate set of z, then hypercube geodesicity gives

$$A \cap B \subseteq Z \subseteq A \cup B.$$

Thus $0 < |Z| < d$. Both pairs (u, z) and (z, v) have ambient distance smaller than d, so minimality yields

$$d_0(u, z) = |Z|, \qquad d_0(z, v) = d - |Z|.$$

Consequently $d_0(u, v) \leq d$, a contradiction. Hence one of the two arcs of C_0 is geodesic for every pair of its vertices, and C_0 is isometric.

Lift C_0 to $\widetilde{G}$ by following P_1 in the negative layer, crossing the e-edge at v, following P_2 backwards in the positive layer, and crossing the e-edge at u. Since C_0 is isometric, if the first half has labels $r_1, \dots, r_d$, the lifted cycle C has label word

$$r_1 \cdots r_d e\, r_1 \cdots r_d e,$$

and is therefore isometric.

Put $M = \mathrm{gh}_{\widetilde{G}}(C)$ and let

$$Z := \{z \in V(G_0) : z^-, z^+ \in V(M)\}.$$

We show that Z is connected. Let p_M denote the gate map onto M. By Theorem 2.3, adjacent vertices have equal or adjacent gates. Since M crosses e, Theorem 2.4 also shows that p_M preserves the e-coordinate.

For $z \in G_0$, the adjacent twins z^-, z^+ therefore have gates that are distinct and adjacent across e. Thus there is a unique $\rho(z) \in Z$ such that

$$p_M(z^-) = \rho(z)^-, \qquad p_M(z^+) = \rho(z)^+.$$

If $z \in Z$, then $\rho(z) = z$. If zz' is an edge of G_0, then $z^-z'^-$ is an edge of the expansion, so $\rho(z)$ and $\rho(z')$ are equal or adjacent. Hence $\rho : G_0 \to Z$ is a graph retraction. Since G_0 is connected, Z is connected.

The vertices u, v belong to Z. Since $Z \subseteq G_0$,

$$d_Z(u, v) \geq d_0(u, v) > d_G(u, v).$$

Thus the connected induced subgraph Z of the ambient hypercube is not isometric. By the virtual-isometric-tree characterization [5, Proposition 9], a connected induced subgraph of a hypercube with VC-dimension at most one is an isometric tree. Therefore $\mathrm{VCdim}(Z) \geq 2$. Choose a pair of coordinates shattered by Z. For every $z \in Z$, the gated hull M contains both z^- and z^+, so adjoining the new coordinate e shows that M shatters a triple. Hence $\mathrm{VCdim}(M) \geq 3$. □

Corollary 8.2 (Connected high-dimensional hyperplanes are impossible)**.** *Let H be a finite partial cube satisfying the gated-hull hypothesis of Conjecture 1.1. Then every connected Θ-hyperplane of H has VC-dimension at most one.*

Proof. For a Θ-class E_e, contract E_e. The original graph is the isometric expansion of the contraction whose overlap is the e-hyperplane. If that overlap were connected and had VC-dimension at least two, Theorem 8.1 would provide an isometric cycle with non-two-dimensional gated hull. □

Corollary 8.3 (Disconnected-hyperplane reduction)**.** *If Conjecture 1.1 has a finite counterexample H, then H has a disconnected Θ-hyperplane of VC-dimension at least two. In fact, every hyperplane of H having VC-dimension at least two is disconnected.*

Proof. Chepoi, Knauer, and Philibert proved that a partial cube has VC-dimension at most two if and only if every hyperplane has VC-dimension at most one [5, Proposition 6 and Corollary 7]. Thus a non-two-dimensional H has a hyperplane of VC-dimension at least two. Such a hyperplane cannot be connected by Theorem 8.2. □

The last corollary reduces a hypothetical obstruction to a disconnected high-dimensional hyperplane: high dimension would have to be distributed across distinct components. The rank-two resolution in Section 10 will remove this remaining possibility.

9. Paired chains and universal trace forcing

We now replace the clean-interval construction by an exact description of all cycles crossing a fixed expansion class. Throughout this section, let

$$H = G^e$$

be the isometric expansion of the partial cube G with respect to an isometric cover (G_1, G_0, G_2). We write $x^{(1)} = x^-$, $x^{(2)} = x^+$, and $\bar{\sigma} = 3 - \sigma$ for $\sigma \in \{1, 2\}$.

Fix distinct $u, v \in V(G_0)$ and put $S = \Delta(u, v)$. For $\sigma \in \{1, 2\}$ define the *paired Boolean region*

$$\mathscr{R}^{\sigma}_{u,v} = \{A \subseteq S : u \triangle A \in V(G_\sigma),\ v \triangle A \in V(G_{\bar{\sigma}})\}. \tag{3}$$

A *paired chain* is a saturated chain

$$\varnothing = A_0 \subset A_1 \subset \cdots \subset A_d = S, \qquad |A_i| = i,$$

contained in $\mathscr{R}^{\sigma}_{u,v}$.

Theorem 9.1 (Exact paired-chain criterion)**.** *Let $d = |S|$.*

(i) *Every paired chain in $\mathscr{R}^{\sigma}_{u,v}$ yields the isometric cycle*

$$C = ((u\triangle A_0)^{(\sigma)}, \dots, (u\triangle A_d)^{(\sigma)}, (v\triangle A_0)^{(\bar\sigma)}, \dots, (v\triangle A_d)^{(\bar\sigma)}) \tag{4}$$

in H. It has length $2d+2$, and its two e-edges have bases u and v.

(ii) *Conversely, every isometric cycle of H crossing the class e arises in this way, up to reversing the cycle and interchanging the two layers.*

Proof. Write $A_i = A_{i-1} \cup \{q_i\}$. Consecutive vertices in the first and second blocks of (4) differ in coordinate q_i. Since $u\triangle S = v$, the two blocks are joined by the e-edges over v and u. The label word of C is

$$q_1 q_2 \cdots q_d \, e \, q_1 q_2 \cdots q_d \, e.$$

Every cyclic subword of length at most $d+1$ has pairwise distinct labels and is therefore geodesic. Hence C is isometric.

Conversely, apply Theorem 2.2. If the cycle crosses e, then e occurs exactly twice. Start immediately after one e-edge and orient the cycle so that the two occurrences of e are the last labels of the two equal half-words. Removing e leaves a repetition-free word $q_1 \cdots q_d$ with $\{q_1, \dots, q_d\} = S$. The successive prefix sets form a saturated chain in one of the regions (3). □

The former clean-interval lemma is now a construction theorem for paired chains.

Corollary 9.2 (Clean-interval construction)**.** *Suppose $u, v \in V(G_0)$ and*

$$I_G(u,v) \cap V(G_0) = \{u, v\}. \tag{5}$$

Then one of the regions $\mathscr{R}^{\sigma}_{u,v}$ contains a paired chain. Equivalently, H contains an e-crossing isometric cycle with bases u, v.

Proof. If $d_G(u,v) = 1$, the four vertices u^-, v^-, v^+, u^+ form an isometric square, and the assertion follows. Assume $d_G(u,v) \geq 2$. Choose u–v geodesics $P_i \subseteq G_i$. Their internal vertices avoid G_0 by (5). A common internal vertex would belong to $G_1 \cap G_2 = G_0$, so the paths have no common internal vertex.

Translate u to $\varnothing$ and write $v = S$. Every vertex of either geodesic is then a subset of S. Let $x = A \in P_1$ and $y = B \in P_2$. If $A \cap B = \varnothing$, the x–y arc through u has length $|A| + |B| = |A\triangle B|$ and is geodesic. If $A \cup B = S$, the arc through v has length $2|S| - |A| - |B| = |A\triangle B|$ and is geodesic.

Suppose neither condition holds. Every x–y path meets G_0, because an edge from $G_1 \setminus G_0$ to $G_2 \setminus G_0$ would belong to neither member of the isometric cover. Choose an x–y geodesic and a vertex $z \in G_0$ on it. By the Hamming interval description,

$$A \cap B \subseteq z \subseteq A \cup B.$$

Thus z is a nonempty proper subset of S, so $z \in I_G(u,v) \cap G_0 \setminus \{u,v\}$, contradicting (5). Hence for every pair of vertices on different paths one of the two cycle arcs is geodesic. Pairs on one path are joined geodesically by a subpath. Therefore $P_1 \cup P_2$ is an isometric cycle. Its standard lift to the two expansion layers is an e-crossing isometric cycle, and Theorem 9.1 supplies the paired chain. □

In particular, a cross-component pair of minimum distance is interval-clean. The crucial point is that Theorem 9.1 also permits non-clean pairs and pairs lying in the same component.

Let C be any e-crossing isometric cycle, let u, v be the bases of its e-edges, and set

$$S = \Delta(u,v), \qquad M = \mathrm{gh}_H(C), \qquad Z_C = \{z \in V(G_0) : z^-, z^+ \in V(M)\}.$$

Theorem 9.3 (Universal trace retraction)**.** *The gate map onto M induces a graph retraction*

$$\rho_C : G_0 \longrightarrow G_0[Z_C]$$

with the following properties:

(i) $\pi_S(\rho_C(x)) = \pi_S(x)$ *for every* $x \in G_0$*; consequently*

$$\pi_S(Z_C) = \pi_S(G_0).$$

(ii) *If* $x, y \in Z_C$*, then*

$$I_G(x,y) \cap V(G_0) \subseteq Z_C.$$

(iii) *If* X *is shattered by* Z_C*, then* $X \cup \{e\}$ *is shattered by* M*. In particular,*

$$\mathrm{VCdim}(M) \geq 1 + \mathrm{VCdim}(Z_C).$$

Proof. Let p_M be the gate map. Because M crosses e, Lemma 2.4 gives different e-coordinates for $p_M(x^-)$ and $p_M(x^+)$. The two images are at distance at most one, so they are the endpoints of a unique e-edge. Define $\rho_C(x)$ by

$$p_M(x^-) = \rho_C(x)^-, \qquad p_M(x^+) = \rho_C(x)^+.$$

If $x \in Z_C$, both copies already lie in M, so $\rho_C(x) = x$. Since a gate map sends adjacent vertices to equal or adjacent vertices, ρ_C is a graph retraction.

The cycle C, and hence M, crosses every coordinate in S. Lemma 2.4 therefore gives $\pi_S(\rho_C(x)) = \pi_S(x)$, proving (i).

For $x, y \in Z_C$ and $z \in I_G(x,y) \cap G_0$, the gate identity with $x^-, y^- \in M$ gives

$$\rho_C(z) \in I_G(z,x) \cap I_G(z,y) = \{z\}.$$

Thus $z \in Z_C$, proving (ii). Finally, M contains both z^- and z^+ for every $z \in Z_C$, which proves (iii). □

We next extract a canonical subset of Z_C without computing the gated hull. For $B \subseteq F := V(G_0)$ and $x \in F$, define

$$\mathcal{P}_{S,B}(x) = \{y \in F : \pi_S(y) = \pi_S(x) \text{ and } y \in I_G(x,z) \text{ for every } z \in B\}. \tag{6}$$

Put

$$\Psi_S(B) = B \cup \{x \in F : \mathcal{P}_{S,B}(x) = \{x\}\}, \tag{7}$$

and, starting with $B_0 = \{u, v\}$, define

$$T_F(u,v) = \bigcup_{j \geq 0} B_j, \qquad B_{j+1} = \Psi_S(B_j). \tag{8}$$

We call $T_F(u,v)$ the *trace-forcing closure* of the pair (u,v).

Theorem 9.4 (Universal trace forcing)**.** *For every e-crossing isometric cycle C with e-edge bases u, v,*

$$T_F(u,v) \subseteq Z_C.$$

Consequently,

$$\mathrm{VCdim}(\mathrm{gh}_H(C)) \geq 1 + \max\{\mathrm{VCdim}(\pi_S(F)), \mathrm{VCdim}(T_F(u,v))\}. \tag{9}$$

Proof. Suppose $B \subseteq Z_C$. For $x \in F$ and $z \in B$, the gate identity applied in either layer gives

$$\rho_C(x) \in I_G(x,z).$$

Theorem 9.3(i) also gives $\pi_S(\rho_C(x)) = \pi_S(x)$. Hence

$$\rho_C(x) \in \mathcal{P}_{S,B}(x).$$

If $\mathcal{P}_{S,B}(x) = \{x\}$, then $\rho_C(x) = x$, so $x \in Z_C$. Since $\{u,v\} \subseteq Z_C$, induction on (8) proves $T_F(u,v) \subseteq Z_C$. The lower bound follows from Theorem 9.3 and $\pi_S(Z_C) = \pi_S(F)$. □

The preceding theorem gives a three-level certificate hierarchy. The projection level uses $\mathrm{VCdim}(\pi_S(F)) \geq 2$. If projection is insufficient, put

$$J_F(u,v) = I_G(u,v) \cap F.$$

Corollary 9.5 (Interval and injective-projection criteria)**.** *For every $u, v \in F$,*

$$J_F(u,v) \subseteq T_F(u,v).$$

Moreover, if π_S is injective on F, then $T_F(u,v) = F$. Hence, once a paired chain exists, each of the following is sufficient for a non-two-dimensional hull:

$$\mathrm{VCdim}(\pi_S(F)) \geq 2, \qquad \mathrm{VCdim}(J_F(u,v)) \geq 2, \qquad \mathrm{VCdim}(T_F(u,v)) \geq 2.$$

Proof. The stable set $T = T_F(u,v)$ is relatively interval-convex in F. Indeed, if $a, b \in T$ and $x \in I_G(a,b) \cap F$, then every $y \in \mathcal{P}_{S,T}(x)$ lies in

$$I_G(x,a) \cap I_G(x,b) = \{x\},$$

so $x \in T$. This proves $J_F(u,v) \subseteq T$. If π_S is injective on F, every prediction set in the first forcing step is the singleton containing its base vertex, so $T = F$. □

For comparison with Section 7, orient every coordinate outside S by its common value on u, v and put

$$A_S(x) = \{q \notin S : x_q \neq u_q\}.$$

Let $L_F(u,v)$ be the union, over all nonempty S-fibres, of the vertices whose A_S-supports are inclusion-minimal in that fibre, and let $K_F(u,v)$ be the relative interval closure of $L_F(u,v)$ in F. The definitions and Corollary 9.5 give

$$L_F(u,v) \subseteq K_F(u,v) \subseteq T_F(u,v) \subseteq Z_C. \tag{10}$$

Thus trace forcing strictly extends both the one-round minimal-fibre rule and its relative interval closure.

10. Rank-two resolution and the overlap Hamming geometry

We now resolve the disconnected-overlap problem. Throughout this section,

$$H = G^e$$

is an isometric expansion of a finite partial cube G, with isometric cover (G_1, G_0, G_2), new coordinate e, and overlap vertex family $F = V(G_0)$. We argue under the temporary hypothesis

$$\mathrm{VCdim}(\mathrm{gh}_H(C)) \leq 2 \quad \text{for every isometric cycle } C \subseteq H. \tag{11}$$

The aim is to show that (11) forces $\mathrm{VCdim}(F) \leq 1$.

10.1. **Binary crossing cells and disk resolution.**

We shall repeatedly use the following direct consequence of the cycle-hull theorem of Chepoi–Knauer–Philibert.

Lemma 10.1 (Rank-two cycle hulls)**.** *Let C be an isometric cycle in a finite partial cube K and put $M = \mathrm{gh}_K(C)$. If $\mathrm{VCdim}(M) \leq 2$, then M is a disk or a full subdivision of a complete graph.*

Proof. The graph M is a two-dimensional partial cube. By locality of gated hulls, $\mathrm{gh}_M(C) = M$. Theorems 33 and 45 of [5] therefore identify M as one of the two stated rank-two cell types. □

An e-crossing isometric cycle has exactly two e-edges. If their bases are $x, y \in F$, its old support is $B = \Delta(x,y)$. We call such a cycle *binary* if

$$\pi_B(F) = \{\pi_B(x), \pi_B(y)\}. \tag{12}$$

Lemma 10.2 (Gated crossing cycles are binary)**.** *Let D be a gated isometric cycle crossing e, with e-edge bases x, y and old support $B = \Delta(x, y)$. Then* (12) *holds.*

Proof. Since D is gated, $\mathrm{gh}_H(D) = D$. Its exact double-layer trace consists of its two e-edge bases x, y. Apply the universal trace retraction Theorem 9.3(i): it preserves every coordinate of B and maps F onto this two-vertex trace. Hence $\pi_B(F) = \pi_B(\{x, y\})$. $\square$

Lemma 10.3 (Disjoint-or-equal binary supports)**.** *If two gated binary crossing cycles have old supports B_1, B_2, then*

$$B_1 \cap B_2 \neq \varnothing \quad \Longrightarrow \quad B_1 = B_2.$$

Proof. Let u, v be the bases of the second cycle. If $B_1 \cap B_2 \neq \varnothing$, then u, v have different B_1-projections. Binaryness of the first cycle says that F has only two, complementary, B_1-patterns. Thus u, v differ on every coordinate of B_1, and $B_1 \subseteq B_2$. Interchanging the two cycles gives the reverse inclusion. $\square$

The next lemma records the rank-two structure of a disk. We use the standard representation of a disk as the region graph of a pseudoline arrangement. Faces of the associated affine oriented matroid are gated; see [5, 11].

Lemma 10.4 (Resolution of a disk segment)**.** *Let M be a gated disk in H crossing e, and let $x, y \in F$ be the bases of two e-edges of M. Then there are bases*

$$x = z_0, z_1, \ldots, z_t = y$$

and gated binary e-crossing cycles $D_1, \ldots, D_t \subseteq M$ such that D_i has e-edge bases z_{i-1}, z_i. If $B_i = \Delta(z_{i-1}, z_i)$, then

$$\Delta(x, y) = B_1 \dot\cup \cdots \dot\cup B_t. \tag{13}$$

Proof. Represent M by a pseudoline arrangement and let ℓ_e be the pseudoline corresponding to e. Along the segment of ℓ_e between the chosen e-edges, list the successive intersection points. At one such point, let B_i be the set of old pseudolines passing through it. The regions incident with that point form the tope graph of a rank-two face, hence a gated even cycle in the disk M. Since M is gated in H, gatedness is transitive and this cycle D_i is gated in H as well. It crosses e, and Theorem 10.2 shows that it is binary on B_i.

A pseudoline meets ℓ_e exactly once, so the sets B_i are pairwise disjoint. Moving past the ith intersection complements precisely the coordinates in B_i. The cumulative change from the first selected e-edge to the second is therefore their disjoint union, proving (13). $\square$

10.2. **Full subdivisions and their trace blocks.** We write SK_n for the graph obtained by subdividing every edge of K_n once. The case $n = 3$ is an even cycle and is already included in the disk resolution. We shall use the standard embedding of SK_n in Q_n, whose original vertices are the singletons $\{i\}$ and whose subdivision vertices are the pairs $\{i, j\}$.

Lemma 10.5 (Standard embedding of a full subdivision)**.** *Let $M \cong \mathrm{SK}_n$, $n \geq 4$, be an isometric subgraph of a partial cube. After permuting and reorienting ambient coordinates, the embedding of M is the standard singleton–pair embedding.*

Proof. This is the standard-embedding lemma [5, Lemma 19]; we include the argument because the precise coordinate model is used below. Choose an original vertex b of M as the zero vector. Its $n - 1$ neighbouring subdivision vertices have pairwise different labels, so after permuting coordinates they are $v_i = \{i\}$, $1 \leq i \leq n - 1$.

Let u_i be the other original endpoint adjacent to v_i. Since $d(b, u_i) = 2$, its label is $\{i, a_i\}$. The second coordinate a_i cannot be one of $1, \ldots, n-1$ different from i: if $a_i = j$, then the vertex $v_j = \{j\}$ would be adjacent to u_i as well as to b and u_j, whereas a subdivision vertex of SK_n has degree two. Because an isometric subgraph of a hypercube is induced, this is impossible. For $i \neq j$ the original vertices u_i, u_j have distance two, so their two-element labels intersect in one coordinate. The labels cannot intersect in i or j by the preceding observation; hence $a_i = a_j$. Denote the common coordinate by n.

Reorient coordinate n. Then b has label $\{n\}$, each u_i has label $\{i\}$, and each v_i has label $\{i, n\}$. Finally, the subdivision vertex between u_i and u_j is a common neighbour of the singletons $\{i\}, \{j\}$. The two possible cube neighbours are $\varnothing$ and $\{i, j\}$. It cannot be $\varnothing$, because that vertex would be adjacent to all $n-1 \geq 3$ singletons and would not have degree two in the induced copy of SK_n. Therefore its label is $\{i, j\}$. This is exactly the standard embedding. □

Let $M \cong \mathrm{SK}_n$, $n \geq 4$, be a gated subgraph of H crossing e, and put

$$R_M = \Theta(M) \setminus \{e\}, \qquad Z_M = \{z \in F : z^-, z^+ \in M\}.$$

In the standard model, the e-edges are indexed by R_M; after a translation there is a vector c_M such that their bases are $c_M \triangle \{p\}$, $p \in R_M$. The halfspace containing the original vertex indexed by e induces a star on these e-edge endpoints; we call it the *star side* of M. Its centre has old-coordinate pattern $c_M|_{R_M}$.

Lemma 10.6 (Full-subdivision trace normal form)**.** *The gate map onto M induces a retraction*

$$\rho_M : F \longrightarrow Z_M$$

that preserves every coordinate of R_M. Consequently

$$\pi_{R_M}(F) = \{\pi_{R_M}(c_M \triangle \{p\}) : p \in R_M\}. \tag{14}$$

In particular, Z_M is independent and ρ_M is constant on every connected component of G_0.

Proof. Let p_M be the gate map onto M. For $x \in F$, the two expansion copies x^-, x^+ are adjacent and have different e-coordinates. Since M crosses e, Theorem 2.4 shows that their gates still have different e-coordinates; by Theorem 2.3 those gates are the endpoints of one e-edge of M. There is therefore a unique $\rho_M(x) \in Z_M$ such that

$$p_M(x^-) = \rho_M(x)^-, \qquad p_M(x^+) = \rho_M(x)^+.$$

The map fixes Z_M, and an edge of G_0 is sent to an edge or a vertex, so ρ_M is a graph retraction.

The subgraph M crosses every coordinate of R_M. Applying Theorem 2.4 in either expansion layer gives $\pi_{R_M}(\rho_M(x)) = \pi_{R_M}(x)$ for all $x \in F$. Since ρ_M maps F onto Z_M, whose standard-coordinate patterns are precisely the translated singletons, this proves (14).

Two distinct trace bases differ in two R_M-coordinates and are not adjacent. Hence the image under ρ_M of each edge of G_0 is a single vertex. It follows that ρ_M is constant on every connected component of G_0. □

Theorem 10.7 (Full-subdivision block dichotomy)**.** *Let M, N be gated crossing full subdivisions, with old coordinate blocks $R = R_M$ and $S = R_N$. Exactly one of the following holds.*

(i) *$R = S$, and the two translated singleton systems in* (14) *agree on this block;*
(ii) *$R \cap S = \varnothing$. In this case all trace bases of N lie in one M-trace fibre, and all trace bases of M lie in one N-trace fibre.*

Thus distinct full-subdivision blocks are pairwise disjoint.

Proof. Regard c_M and c_N as the ambient old-coordinate patterns shared by the trace bases of the respective cells outside their old blocks. Translate the R-coordinates by c_M and put $a = (c_N \triangle c_M)|_R$. For every $s \in S$, the s-trace base of N belongs to F, so its R-projection must be a singleton. Equivalently,

$$a \triangle (\{s\} \cap R) \quad \text{is a singleton subset of } R \qquad (s \in S). \tag{15}$$

If some $s \in S \setminus R$ exists, then a is a singleton. For any $s \in S \cap R$, however, toggling s makes it empty or a two-set, contrary to (15). Hence $S \cap R = \varnothing$, and all N-leaves have one common R-projection. The symmetric argument gives case (ii).

Otherwise $S \subseteq R$. If some $s \in S$ is not in a, then (15) forces $a = \varnothing$. If every $s \in S$ lies in a, the same condition forces $|a| = 2$, impossible because $|S| = n - 1 \geq 3$. Thus $a = \varnothing$ and $S \subseteq R$; symmetry gives $R = S$, proving case (i). □

Corollary 10.8 (Parallel star-side orientation)**.** *Two full subdivisions with the same old block R have their star sides in the same e-layer.*

Proof. Their virtual centre patterns agree on R and may differ only on coordinates outside R. In the standard model, the star centre has the virtual centre pattern on R. Suppose the two actual star centres lie in opposite e-layers. Their endpoints agree on every coordinate of R. Hence every Hamming geodesic between them, and in particular every graph geodesic, keeps all R-coordinates fixed. Such a geodesic must cross the class e at an e-edge whose base lies in F and has the virtual centre pattern on R. Equation (14) says that every F-pattern on R is a translated singleton, never the centre pattern. This contradiction proves the claim. □

Lemma 10.9 (Binary transversality)**.** *Let D be a gated binary crossing cycle, with bases x, y and support $B = \Delta(x, y)$. For every gated crossing full subdivision M,*

$$\rho_M(x) = \rho_M(y), \qquad B \cap R_M = \varnothing.$$

Proof. If the two trace labels were distinct, say $p, q \in R_M$, then $p, q \in \Delta(x, y) = B$. Choose a third label $r \in R_M \setminus \{p, q\}$. On the coordinates p, q, the three trace bases have, after translation, the patterns $10, 01, 00$. The r-trace base therefore has a B-projection different from both endpoint projections, contrary to binaryness. Thus the two labels agree. Since ρ_M preserves R_M, the bases agree on R_M, proving the second assertion. □

10.3. **The resolved state graph.** Let $\mathscr{R}$ be the set of distinct full-subdivision blocks. By Theorem 10.7, its members are pairwise disjoint. For $R \in \mathscr{R}$, choose its translated centre pattern c_R and define the state

$$\lambda_R(x) = p \quad \Longleftrightarrow \quad \pi_R(x) = \pi_R(c_R \triangle \{p\}) \qquad (x \in F).$$

Let $\mathscr{B}$ be the set of distinct supports of gated binary crossing cycles. By Theorems 10.3 and 10.9, the blocks in $\mathscr{R} \cup \mathscr{B}$ are pairwise disjoint. A binary block has two complementary F-patterns; denote the corresponding bit by $\lambda_B(x) \in \{0, 1\}$.

Define the Hamming graph (equivalently, a Cartesian product of complete graphs; see [17])

$$\mathcal{H}_e = \prod_{R \in \mathscr{R}} K_{|R|} \,\square \prod_{B \in \mathscr{B}} K_2 \tag{16}$$

and the state map

$$\lambda : F \longrightarrow V(\mathcal{H}_e), \qquad \lambda(x) = ((\lambda_R(x))_{R \in \mathscr{R}}, (\lambda_B(x))_{B \in \mathscr{B}}).$$

The *resolved transition graph* Γ_e has vertex set F. Two vertices are adjacent in Γ_e if either

(a) they are the bases of a gated binary crossing cycle; or
(b) they are two trace bases of one gated crossing full subdivision.

Such an edge changes exactly one factor of (16).

Theorem 10.10 (Resolved Hamming geometry)**.** *Under hypothesis* (11)*, the graph Γ_e is connected and*

$$\lambda : \Gamma_e \longrightarrow \mathcal{H}_e$$

is an isometric embedding.

Proof. Fix $x, y \in F$. Translate coordinates so that $x = \varnothing$ and write $y = \Delta(x, y)$. The set $F \cap I_G(x, y)$ is a finite subposet of the Boolean interval $[\varnothing, y]$. Choose a maximal inclusion chain

$$x = x_0 \subset x_1 \subset \cdots \subset x_t = y. \tag{17}$$

If a vertex of F lay strictly between two consecutive terms, it could be inserted in the chain. Hence

$$I_G(x_{i-1}, x_i) \cap F = \{x_{i-1}, x_i\}.$$

By Theorem 9.2, there is an e-crossing isometric cycle with these bases. Its gated hull has VC-dimension at most two by (11), so Theorem 10.1 makes it a disk or a full subdivision.

In the disk case, Theorem 10.4 replaces the step by a path of binary transitions whose pairwise disjoint supports partition $\Delta(x_{i-1}, x_i)$. In the full-subdivision case, the two bases are two distinct trace leaves. Their translated singleton patterns differ in two coordinates $p, q \in R$, and the corresponding edge of the complete factor $K_{|R|}$ changes the state from p to q.

The increments $\Delta(x_{i-1}, x_i)$ of the inclusion chain are pairwise disjoint. It follows that no resolved factor changes twice. A second binary transition would reuse every coordinate of the same binary support. For a full-subdivision factor, suppose a first transition changes the state p to q. Until this factor is changed again its state remains q; a second transition therefore starts at q and has coordinate increment $\{q, r\}$ for some r, which meets the first increment $\{p, q\}$. Both possibilities contradict disjointness of the chain increments.

Concatenating the resolved pieces gives an x–y path in Γ_e. Every changed factor is changed exactly once, and because the path ends at y, these are precisely the factors on which $\lambda(x)$ and $\lambda(y)$ differ. Therefore

$$d_{\Gamma_e}(x, y) \le d_{\mathcal{H}_e}(\lambda(x), \lambda(y)).$$

Conversely, every edge of Γ_e changes exactly one Hamming factor, so any x–y path has length at least the Hamming distance of the endpoint states. Equality holds. In particular, the construction joins every pair of vertices, and equality for state distance zero also proves that λ is injective. Thus Γ_e is connected and λ is an isometric embedding. ☐

10.4. **Mixed cycles.** Give each edge of Γ_e the colour of its changed Hamming factor. A cycle is *mixed* if it uses at least two colours.

Lemma 10.11 (A shortest mixed cycle is isometric)**.** *A mixed cycle of minimum length in Γ_e is isometric, and no two consecutive edges of it have the same colour.*

Proof. Choose a mixed cycle Q of minimum length. Suppose first that two consecutive edges xy, yz of Q have the same colour. Because $\lambda(\Gamma_e)$ is an isometric subgraph of a Hamming graph, the first and third vertices are adjacent in Γ_e (they differ in at most that one factor). Replacing xy, yz by xz deletes the vertex y and gives a strictly shorter simple cycle. All other edges of Q remain, so a second colour is still present. This contradicts the minimal choice of Q. Thus consecutive colours are distinct.

If Q is not isometric, choose $u, v \in V(Q)$ and a u–v geodesic P whose length is smaller than the lengths of both u–v arcs Q_1, Q_2 of Q. Each closed walk $P \cup Q_i$ is shorter than Q. Decompose its modulo-two edge set into simple cycles. By minimality of Q, every cycle in both decompositions is monochromatic. The two copies of P cancel over $\mathbb{F}_2$, so the sum of the two decompositions is exactly the edge set of Q.

For any fixed colour, a modulo-two sum of monochromatic cycles has even degree at every vertex in the subgraph formed by that colour. At a colour-change vertex of the simple mixed cycle Q, however, each of the two incident colours has degree one in its colour subgraph. This contradiction shows that Q is isometric. ☐

Lemma 10.12 (Factor word)**.** *Let Q be an isometric mixed cycle in a Hamming graph, with no consecutive edges of one colour. Then $|Q| = 2k$, and after cyclic reindexing its factor word is*

$$A_1A_2\cdots A_kA_1A_2\cdots A_k, \tag{18}$$

where $A_1,\dots,A_k$ are pairwise distinct.

Proof. Put $\ell = |Q|$. Every cycle arc of length at most $\lfloor \ell/2\rfloor$ is a geodesic, and a Hamming geodesic changes each factor at most once.

Fix a factor and list the positions of its edges cyclically. Any cycle arc of length at most $\lfloor \ell/2\rfloor$ is a geodesic and hence contains at most one edge of that factor. If the factor occurred at least three times, let g be the number of edges strictly after one occurrence and up to and including the next occurrence, chosen minimum over the cyclically consecutive pairs. The corresponding gaps sum to ℓ, so $g \le \lfloor \ell/3\rfloor$. The arc that starts with the first factor edge and ends with the next has length $g+1$. The absence of consecutive equal colours implies $\ell \ge 6$, and hence $g+1 \le \lfloor \ell/3\rfloor + 1 \le \lfloor \ell/2\rfloor$. This geodesic arc would repeat the factor, a contradiction. Thus every factor occurs at most twice.

A factor cannot occur exactly once in a closed walk, since no other factor can restore its state. Hence every used factor occurs exactly twice, and the second change restores the state created by the first. If the two occurrences were not antipodal, one of the two arcs containing both factor edges would have length at most $\lfloor \ell/2\rfloor$ and would not be a Hamming geodesic. Therefore the occurrences are antipodal, $\ell = 2k$, and each half of the cycle contains every used factor exactly once. Write the two half-words as $A_1,\dots,A_k$ and $B_1,\dots,B_k$.

For $0 \le r \le k$, let $P_r = \{A_1,\dots,A_r\}$ and $Q_r = \{B_1,\dots,B_r\}$. Start at a vertex v, and let v' be the vertex reached after the first half. The first occurrence of each factor changes its state away from the state at v; its second occurrence must restore that initial state. Therefore the vertex reached after r edges of the first half and the vertex reached after r edges of the second half differ in

$$|P_r \cap Q_r| + k - |P_r \cup Q_r| = k - |P_r \triangle Q_r|$$

factors. These two vertices are antipodal on the cycle, so both cycle arcs between them have length k. Isometry forces their Hamming distance to be k, hence $P_r = Q_r$ for every r. Comparing successive prefix sets gives $A_r = B_r$ for all r, and the factor word is $A_1\cdots A_kA_1\cdots A_k$. □

The synchronization needed to lift (18) is encoded by gates.

Lemma 10.13 (Synchronization of equal binary blocks)**.** *Let D_1, D_2 be gated binary crossing cycles with the same support B, and suppose their two endpoint patterns on B are matched. If one cycle has a side word w, the gated hull of the other contains a crossing cycle with the same side word and the matched bases.*

Proof. Let p be the gate map onto the second cycle. The target cycle crosses e and every coordinate of B, so Theorem 2.4 implies that p preserves every label occurring on the first cycle. If xy is one of its edges with label $q \in B \cup \{e\}$, then $p(x)$ and $p(y)$ still differ in q; nonexpansiveness gives distance at most one, so they are adjacent and the image edge has label q. Thus no edge is collapsed, and the image is a closed walk with label word $wewe$. By Theorem 2.2, it is a simple isometric cycle (indeed it must be the whole target cycle).

The exact e-trace of the target cycle consists of its two e-edge bases. The gate map preserves the B-projection, and those bases have the two distinct patterns in $\pi_B(F)$. The prescribed endpoint matching therefore uniquely determines the images of the two bases. The image cycle has the required bases and side word. □

Theorem 10.14 (Mixed-cycle concentration)**.** *Under hypothesis* (11)*, the graph Γ_e contains no mixed cycle.*

Proof. Suppose that a mixed cycle exists and choose one of minimum length. By Theorems 10.11 and 10.12, its factor word is

$$A_1 \cdots A_k A_1 \cdots A_k,$$

where the first-half factors are pairwise distinct. Write its consecutive bases as $z_0, z_1, \ldots, z_{2k} = z_0$.

For the first-half transition $z_{i-1}z_i$, choose a connector in the negative expansion layer. If A_i is binary, use the negative side of a gated binary crossing cycle; its label word is a repetition-free ordering w_i of the support A_i. If $A_i = R$ is a full-subdivision factor and the state changes from p to q, use the two-edge path in the appropriate side of the standard SK_n. If the non-star side is negative, the negative p–q path has labels q, p, while the positive q–p path through the star centre also has labels q, p. If the star side is negative, both words are p, q. In either orientation denote the common ordered pair by w_i.

At the antipodal occurrence of A_i, the factor returns from the state of z_i to the state of z_{i-1}. For a binary factor, Theorem 10.13 allows the corresponding positive-layer connector to be chosen with the same word w_i. For a full-subdivision factor, Theorem 10.7 makes the two cells parallel, and Theorem 10.8 puts their star sides in the same layer; the positive connector in the reverse state direction consequently has the same two-letter word w_i.

Concatenate the first k negative connectors from z_0 to z_k, cross the e-edge over z_k, concatenate the k positive connectors back to z_0, and cross the e-edge over z_0. The resulting closed walk has label word

$$WeWe, \qquad W = w_1 w_2 \cdots w_k.$$

The blocks of different factors are disjoint. Each binary word w_i uses every coordinate of its block once, and a full-subdivision word uses two distinct leaf coordinates. Hence W is repetition-free. By Theorem 2.2, the closed walk is an isometric cycle C.

Because the factor cycle is mixed, choose two different factors among the A_i. For a binary factor choose any coordinate in its support. For a full-subdivision transition from state p to state q, choose one of the two old coordinates p, q, which distinguishes those states. Along the balanced factor cycle the two selected coordinates realize the patterns $00, 10, 11, 01$. Every base has both expansion copies, so the environment shatters these two coordinates together with e. All three coordinates occur on C, and Theorem 7.2 yields $\mathrm{VCdim}(\mathrm{gh}_H(C)) \geq 3$, contradicting (11). ☐

10.5. **Eliminating the nonmixed branch.** We finish with a general Hamming-graph lemma.

Lemma 10.15 (Convex factor fibres)**.** *Let X be an isometric subgraph of a Hamming graph $\prod_{i \in I} K_{n_i}$. For one factor i and any $A \subseteq V(K_{n_i})$, the set*

$$X_A = \{x \in X : x_i \in A\}$$

is convex in X.

Proof. A Hamming geodesic changes each factor at most once. If its endpoint states in factor i belong to A, that factor is either unchanged or moves directly between two states of A. Thus the corresponding subset of the full Hamming graph is convex. Isometry of X proves the claim. ☐

Lemma 10.16 (Crossing factor splits force a mixed cycle)**.** *Let $\alpha, \beta : V(X) \to \{0,1\}$ be nonconstant functions depending on two distinct Hamming factors. If all four sets*

$$X_{rs} = \{x : \alpha(x) = r,\ \beta(x) = s\}, \qquad r, s \in \{0,1\},$$

are nonempty, then X contains a mixed cycle.

Proof. The two sides of each factor split are convex by Theorem 10.15; their intersections X_{rs} are therefore convex and, when nonempty, connected. In the convex set $\alpha^{-1}(0)$ choose a geodesic whose endpoints lie in X_{00} and X_{01}. The endpoint values of β differ. A Hamming geodesic changes the factor controlling β exactly once, so this path contains an edge

$$e_0 : X_{00} \longleftrightarrow X_{01}$$

of that factor. Repeating the argument in $\alpha^{-1}(1)$ gives $e_1 : X_{10} \leftrightarrow X_{11}$. Similarly, geodesics in $\beta^{-1}(0)$ and $\beta^{-1}(1)$ give edges

$$f_0 : X_{00} \longleftrightarrow X_{10}, \qquad f_1 : X_{01} \longleftrightarrow X_{11}$$

of the factor controlling α.

Inside each connected quadrant join the endpoints of its two incident cross-edges by a simple path, taking a path of length zero when the two endpoints coincide. The four quadrants are pairwise disjoint. Each internal vertex of a chosen path has degree two in the union, and each path endpoint is incident with exactly one internal-path edge (unless the path has length zero) and exactly one cross-edge. Consequently the union of the four paths and the four cross-edges is connected and 2-regular, hence is a simple cycle. It uses both controlling factors and is mixed. □

Lemma 10.17 (A shattered old-coordinate pair uses two factors)**.** *If old coordinates a, b are shattered by F, then they belong to two distinct factors of* (16)*.*

Proof. Every old coordinate that varies on F belongs to some resolved factor: join two bases on which it differs by a path in the connected graph Γ_e and inspect an edge where it changes. Two coordinates in one binary block see only its two complementary patterns. Two coordinates in one full-subdivision block see a translated singleton family and hence at most three of the four Boolean patterns. Neither pair can be shattered. □

Theorem 10.18 (Expansion obstruction)**.** *If* $\mathrm{VCdim}(F) \geq 2$*, then H contains an isometric cycle C satisfying*

$$\mathrm{VCdim}(\mathrm{gh}_H(C)) \geq 3.$$

No connectedness assumption on G_0 is required.

Proof. Assume (11). The resolved Hamming geometry Theorem 10.10 applies, and by Theorem 10.14 the isometric state graph Γ_e has no mixed cycle. Choose old coordinates a, b shattered by F. By Theorem 10.17, they lie in distinct factors. Each coordinate is a binary function of its factor state: in a binary factor it is the factor bit or its complement; in a full-subdivision factor it is the indicator of one singleton state or its complement. Since a, b are shattered, all four intersections of the corresponding binary factor splits are nonempty. Apply Theorem 10.16 to the isometric subgraph $\lambda(\Gamma_e)$ of $\mathcal{H}_e$. It contains a mixed cycle, contradicting Theorem 10.14. □

11. The characterization theorem and consequences

We now deduce the conjectured characterization from the expansion obstruction.

Theorem 11.1 (Gated-hull characterization)**.** *For a finite partial cube K, the following are equivalent.*

(i) $\mathrm{VCdim}(K) \leq 2$*;*
(ii) *the gated hull of every isometric cycle of K is a disk or a full subdivision of a complete graph.*

Proof. The implication (i)⇒(ii) is the theorem of Chepoi, Knauer, and Philibert [5, Theorems 33 and 45].

For the converse, suppose $\mathrm{VCdim}(K) \geq 3$. By the hyperplane characterization of VC-dimension [5, Proposition 6 and Corollary 7], some Θ-hyperplane has VC-dimension at least two. Contract the corresponding Θ-class. The original graph is the associated isometric expansion, and the overlap is the contracted hyperplane. Apply Theorem 10.18. We obtain an isometric cycle C with $\mathrm{VCdim}(\mathrm{gh}_K(C)) \geq 3$. A disk and a full subdivision both have VC-dimension at most two, so condition (ii) fails. □

Corollary 11.2 (Cyclic detection of the 3-cube obstruction). *A finite partial cube has a shattered triple of Θ-classes if and only if some isometric cycle has a gated hull containing a shattered triple.*

Corollary 11.3 (No minimal counterexample). *The structural target described by the antipodal, COM, separator, product, and disconnected-hyperplane reductions is empty: no finite gated-minimal non-two-dimensional partial cube satisfies the local disk/full-subdivision hull condition.*

Remark 11.4 (Logical scope of the proof). The proof of Theorem 11.1 is independent of all finite enumerations. Its external inputs are the standard metric theory of partial cubes and the rank-two cycle-hull and hyperplane theorems of Chepoi–Knauer–Philibert. In particular, no exhaustiveness assumption on a list of partial cubes, no SAT or integer-programming certificate, and no computer-assisted case split enters the argument.

12. Conclusion

We have proved the gated-hull characterization conjectured by Chepoi, Knauer, and Philibert for finite partial cubes. The principal local theorem is the expansion obstruction: an overlap of VC-dimension at least two, connected or not, forces an isometric cycle whose gated hull has VC-dimension at least three.

The proof separates metric forcing from rank-two geometry. Interval forcing, cycle projection, paired chains, and trace retractions identify the information carried by one crossing cycle. The disk/full-subdivision structure theorem then resolves all possible two-dimensional crossing hulls into disjoint coordinate factors. The resulting overlap state graph is an isometric subgraph of a Hamming graph. Mixed state cycles lift to high-dimensional gated cycle hulls, while the absence of mixed cycles is incompatible with a shattered pair of old coordinates. This closes the only remaining expansion case and yields the global characterization.

The argument is purely theoretical and finite. It uses neither an exhaustive list of partial cubes nor a computer-assisted case analysis.

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Use of generative AI. We used ChatGPT to brainstorm the structure, anticipate potential objections, and refine the text. We are fully responsible for all claims, citations, calculations, and the final wording.

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